# Classical Machine-Learning Models for Groundwater Quality Classification in Telangana, India: Irrigation (Salinity–Sodium) and RSC Suitability Classes, 2018–2020

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**Abstract.** Groundwater is a major source of irrigation and drinking water in Telangana, but classifying every sample in the laboratory is slow and costly. This study asks whether classical machine-learning (ML) models from an undergraduate syllabus can reproduce two standard hydrochemical suitability labels from routine water-chemistry measurements. We use 1,106 post-monsoon groundwater records from 33 districts (2018–2020) with 16 chemical features and two targets: a three-class label (P.S., MR, U.S.) whose class boundaries coincide with residual sodium carbonate (RSC) thresholds, and a 13-label irrigation class based on the USSL salinity–sodium system. Models were compared with five-fold stratified group cross-validation in which identical chemical profiles never cross folds, with all preprocessing inside pipelines. For the RSC-based label, a decision tree reached macro-F1 = 0.995 and accuracy = 0.999, but it did so by learning two thresholds on the RSC feature alone; the label is therefore a deterministic function of one input, and this score shows rule recovery rather than general predictive skill. The other four classifiers reached macro-F1 between 0.607 and 0.799. For the 13-label irrigation class, a support-vector machine (SVM) achieved accuracy = 0.902 but macro-F1 = 0.442, because nine of the 13 labels have 12 or fewer records. TDS and SAR were the most influential features for this task. Simple models can therefore reproduce the common classes well, while rare classes and label-derivation effects need careful treatment.

**Keywords:** groundwater quality; Telangana; irrigation suitability; residual sodium carbonate; sodium adsorption ratio; support vector machine; decision tree; grouped cross-validation; target leakage; SDG 6

## 1 Introduction

Groundwater is a major source of irrigation and drinking water in India, and its chemistry decides whether it can be used safely. Two rating systems are widely applied to irrigation water. The US Salinity Laboratory (USSL) diagram combines a salinity class (C1–C4, based on electrical conductivity) with a sodium-hazard class (S1–S4, based on the sodium adsorption ratio, SAR) [1]. The residual sodium carbonate (RSC) rating labels water as probably safe (below 1.25 meq/L), marginal (1.25–2.5 meq/L) or unsuitable (above 2.5 meq/L) [1], [2]. Assigning these labels needs a full laboratory analysis of each sample. If simple models can reproduce the labels from routine measurements, monitoring agencies can screen samples faster and prioritise the ones that need full analysis. This is relevant to SDG 6 (Clean Water and Sanitation) and SDG 2 (Zero Hunger).

Several recent studies apply ML to Telangana groundwater data and report high accuracies [3]–[5]. A recurring question is how much of such accuracy reflects genuine learning and how much reflects the way the labels were produced or the data were split. We therefore ask how well classical models classify groundwater suitability under a leakage-aware protocol, and what the results reveal about the labels themselves. Our contributions are: (i) a grouped cross-validation protocol in which identical chemical profiles never straddle training and validation folds; (ii) a comparison of five classifiers on the three-class RSC-based label using class-wise and balanced metrics, with nested tuning; (iii) an evaluation of a 13-label USSL irrigation class with an SVM, including class-weight experiments and permutation importance; and (iv) a demonstration that the RSC-based label can be recovered from a single feature by two thresholds, with a discussion of what this means for interpreting accuracy.

Section 2 reviews related work, Section 3 describes the data, Sections 4–5 the methods and set-up, Section 6 the results and Section 7 the discussion, followed by conclusions.

## 2 Related Work

Richards [1] defined the USSL salinity–sodium diagram, and Wilcox [2] summarised the RSC categories used for irrigation water; the two label systems in our data follow these traditions. On the ML side, Rozhkova et al. [3] trained a decision tree on Telangana groundwater chemistry for 2018–2020 and reported an accuracy of 95.7%, highlighting sodium, TDS, hardness and a dissolved-salts ratio as influential variables. Olentsova et al. [4] modelled post-monsoon 2020 groundwater quality using data from the Telangana State Groundwater Department. A master's project by Jot [5] compared boosting and stacked ensembles (LightGBM, random forest, ANN) on the 2018–2020 data with attention to class imbalance. Outside Telangana, Jose and Yasala [6] combined a hidden Markov model with an ANN for groundwater quality in Kanyakumari, Tamil Nadu, and reported 97.41% accuracy.

These studies show that high accuracies are attainable. The abstracts we reviewed foreground accuracy, whereas for imbalanced suitability classes balanced and per-class metrics are more informative. Our work differs in using grouped cross-validation and nested tuning, reporting macro-F1 and per-class recall, and testing explicitly whether a label can be recovered from one input. The general problem of target leakage, where an input encodes the label, is described in [7]. All models were implemented with scikit-learn [8]; the algorithms follow [9]. Our numbers are not directly comparable with [3]–[6], which use different targets and validation schemes.

## 3 Dataset and Preprocessing

### 3.1 Source and description

We used three yearly post-monsoon files (ground_water_quality_2018_post.csv, ground_water_quality_2019_post.csv and ground_water_quality_2020_post.csv) with 374, 364 and 368 records. They were concatenated into one table of 1,106 rows. The yearly files have 26, 26 and 27 columns; the merged table has 38 columns because some parameters are stored under different column names in different files (Section 3.3). The records cover 33 districts and 316 mandals of Telangana. There are 374 unique latitude values, about one-third of the number of records, which indicates that most locations were sampled in more than one year. The data appear to correspond to the groundwater-quality records of the Telangana State Groundwater Department used in [3], [4] [Student: confirm the

exact source and add its URL under Data availability]. There are no exact duplicate rows, but 8 chemical profiles occur twice (1,098 unique profiles), and identical profiles never carry conflicting labels. Table 1 lists the variables.

**Table 1.** Variables used from the merged table, with the percentage of missing values.

| Variable | Description | Missing % | Use |
|---|---|---|---|
| pH | Acidity/alkalinity | 0.0 | feature (stored as text; converted to numeric) |
| TDS | Total dissolved solids | 0.0 | Feature |
| E.C | Electrical conductivity | 32.9 | Feature |
| CO3, HCO3 | Carbonate, bicarbonate | 32.9 | Feature |
| Cl, F, NO3, SO4 | Chloride, fluoride, nitrate, sulphate | 32.9 | Feature |
| Na, K, Ca, Mg | Sodium, potassium, calcium, magnesium | 32.9 | Feature |
| T.H | Total hardness | 0.0 | Feature |
| SAR | Sodium adsorption ratio | 0.0 | Feature |
| RSC (meq/L) | Residual sodium carbonate | 0.0 | Feature |
| gwl | Groundwater level | 1.0 | not used |
| Classification | USSL class (13 labels) | 0.0 | target 1 |
| Classification.1 | RSC-based class (P.S., MR, U.S.) | 0.0 | target 2 |

District, mandal, village, coordinates and season were not used as features.

### 3.2 Target labels

The RSC-based label (Classification.1) has three classes: P.S. (982 records, 88.8%), U.S. (75, 6.8%) and MR (49, 4.4%). Following the usual rating [1], [2], these abbreviations appear to stand for probably safe, unsuitable and marginal [Student: confirm against the source's data dictionary]. The notebook also referred to this target as a drinking-water classification, but its class boundaries match the irrigation RSC thresholds (Table 2), so we describe it as an RSC-based suitability class. RSC and SAR are computed from ion concentrations in meq/L as

$$RSC = (CO_3^{2-} + HCO_3^{-}) - (Ca^{2+} + Mg^{2+}),\quad SAR = Na^{+} / \sqrt{[(Ca^{2+} + Mg^{2+}) / 2]}. \quad (1)$$

**Table 2.** RSC (meq/L) by RSC-based class, all 1,106 records.

| Class | n | Min | Max | Mean | Median |
|---|---|---|---|---|---|
| P.S. | 982 | −59.5845 | 1.2450 | −3.0631 | −1.7981 |

| Class | n | Min | Max | Mean | Median |
|---|---|---|---|---|---|
| MR | 49 | 1.3064 | 2.4020 | 1.8016 | 1.6470 |
| U.S. | 75 | 2.5745 | 18.2008 | 5.7193 | 4.2008 |

The three ranges do not overlap, and the gaps fall at 1.25 and 2.5 meq/L. The classes are therefore perfectly separable by RSC alone.

The irrigation label (Classification) has 13 values: C3S1 (696), C2S1 (248), C4S1 (87), C4S2 (36), C3S2 (12), C4S4 (7), C3S3 (6), C4S3 (5), C1S1 (3), O.G (2), OG (2), C3S4 (1) and C2S2 (1). The first part of each code is the salinity class and the second the sodium class [1]. O.G and OG look like two spellings of one label (its meaning is not documented in the notebook), but they were kept separate. The majority class, C3S1, holds 62.9% of the records, and nine labels have 12 or fewer records.

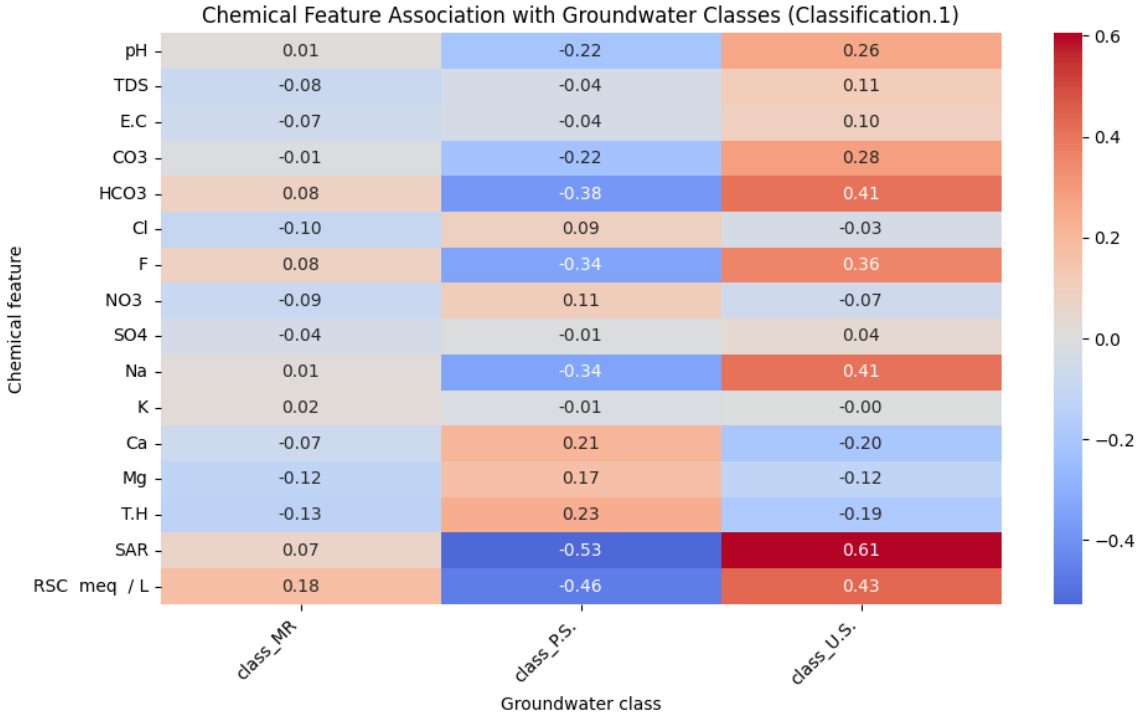
### 3.3 Missing values and data-quality checks

E.C and ten ion variables (CO3 to Mg) are missing for 364 records (32.9%) each; this equals the number of rows in the 2019 file. In the merged table the same parameters also appear under alternative headings (for example EC, Cl −, $HCO_3$ − and Na+) that are non-missing for 364 rows and missing for 742, which suggests that the 2019 file stores them under different names. These duplicate-named columns were not merged in the present analysis, so a third of the records have no value for these 11 features. Missing values were filled with the training-fold median inside the pipeline, so validation data never influenced imputation. pH, TDS, T.H, SAR, RSC and both labels are complete. We did not detect or remove outliers, because extreme chemistry values are real hydrogeological events rather than errors.

### 3.4 Association of features with the labels

Pearson correlations between each feature and the class indicators of the RSC-based label show that SAR has the strongest association (r = 0.605 with U.S. and −0.527 with P.S.), followed by RSC (−0.462 with P.S., 0.435 with U.S., 0.177 with MR), Na (0.408 with U.S.) and HCO3 (0.407 with U.S.).

Fig. 1. Feature–class correlation heatmap for the RSC-based label. Pearson correlation between the 16 chemical features and the class indicators.




## 4 Methodology

All steps after loading were wrapped in scikit-learn Pipelines that were fitted on the training part of each fold only: median imputation, then standardisation (for logistic regression, k-NN, Gaussian Naive Bayes and SVM; tree models were not scaled), then the classifier. Five classifiers were compared on the RSC-based label. **Logistic regression** models class probabilities with a regularised linear model (inverse regularisation strength C, class weights balanced). A **decision tree** chooses splits that minimise Gini impurity. **k-NN** takes a majority vote of the k = 5 nearest standardised neighbours. **Gaussian Naive Bayes** assumes conditionally independent, normally distributed features. The **SVM** uses an RBF kernel and a soft margin controlled by C. For the 13-label irrigation class, only the SVM was evaluated.

Metrics are accuracy, balanced accuracy (the mean of per-class recall), per-class precision, recall and F1, and macro-F1 = $(1/K)\ \Sigma_k\ F1_k$ over the K classes. Because the classes are imbalanced and an unsafe sample classified as safe is the costlier error, macro-F1 and the recall of the unsuitable class (U.S.) are emphasised over accuracy. Feature influence was measured by permutation importance: the mean drop in held-out macro-F1 when one feature is shuffled (10 repeats).

## 5 Experimental Setup

**Validation.** We used StratifiedGroupKFold with 5 folds, shuffling and random seed 42. Records with identical chemical profiles received the same group id (1,098 groups), so a profile never appears in both training and validation data. No separate test set was held out; all estimates are cross-validated. For the 13-label target, several labels have fewer than five records, so some validation folds do not contain every label.

**Tuning.** Hyper-parameters of the SVM, decision tree and logistic regression were tuned by grid search on the grouped folds, selecting by macro-F1. The SVM grid covered $C \in \{1, 10\}$, $\gamma \in \{0.01, \text{scale}\}$ and several class-weight schemes. Because scores from the tuning loop are optimistic, we also ran nested grouped cross-validation, in which tuning is repeated inside each outer training fold.

**Environment.** Python 3.13 in Google Colab with scikit-learn, pandas, NumPy, Matplotlib and seaborn; random seed 42. The notebook is cited under Data availability.

## 6 Results

### 6.1 RSC-based suitability class

A classifier that always predicts P.S. would score 0.888 accuracy but a balanced accuracy of only 0.333. Table 3 shows the baseline comparison.

**Table 3.** RSC-based label: mean ± SD over five grouped folds (untuned baselines).

| Model | Accuracy | Macro-F1 | Balanced accuracy |
|---|---|---|---|
| Decision tree | 0.9991 ± 0.0018 | 0.9954 ± 0.0092 | 0.9917 ± 0.0167 |
| Logistic regression | 0.9322 ± 0.0215 | 0.7987 ± 0.0625 | 0.8841 ± 0.0540 |
| Gaussian Naive Bayes | 0.8987 ± 0.0109 | 0.7158 ± 0.0300 | 0.8284 ± 0.0357 |
| SVM (RBF) | 0.8970 ± 0.0306 | 0.7144 ± 0.0910 | 0.8139 ± 0.0938 |
| k-NN (k = 5) | 0.9285 ± 0.0140 | 0.6070 ± 0.0815 | 0.5628 ± 0.0721 |

The decision tree is almost perfect. k-NN has the second-highest accuracy but a balanced accuracy of 0.563, which shows how accuracy hides poor minority-class performance. Table 4 gives per-class results from pooled out-of-fold predictions.

**Table 4.** Per-class precision (P), recall (R) and F1 for the two minority classes (pooled out-of-fold predictions).

| Model | U.S. P | U.S. R | U.S. F1 | MR P | MR R | MR F1 |
|---|---|---|---|---|---|---|
| Logistic regression | 0.9412 | 0.8533 | 0.8951 | 0.3818 | 0.8571 | 0.5283 |
| Decision tree | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9796 | 0.9897 |
| k-NN | 0.9362 | 0.5867 | 0.7213 | 0.3333 | 0.1020 | 0.1562 |
| Gaussian Naive Bayes | 0.5699 | 0.7067 | 0.6310 | 0.4286 | 0.8571 | 0.5714 |
| SVM (RBF) | 0.8929 | 0.6667 | 0.7634 | 0.3066 | 0.8571 | 0.4516 |

Fig. 2. Grouped-CV confusion matrices of the five classifiers for the RSC-based suitability class.

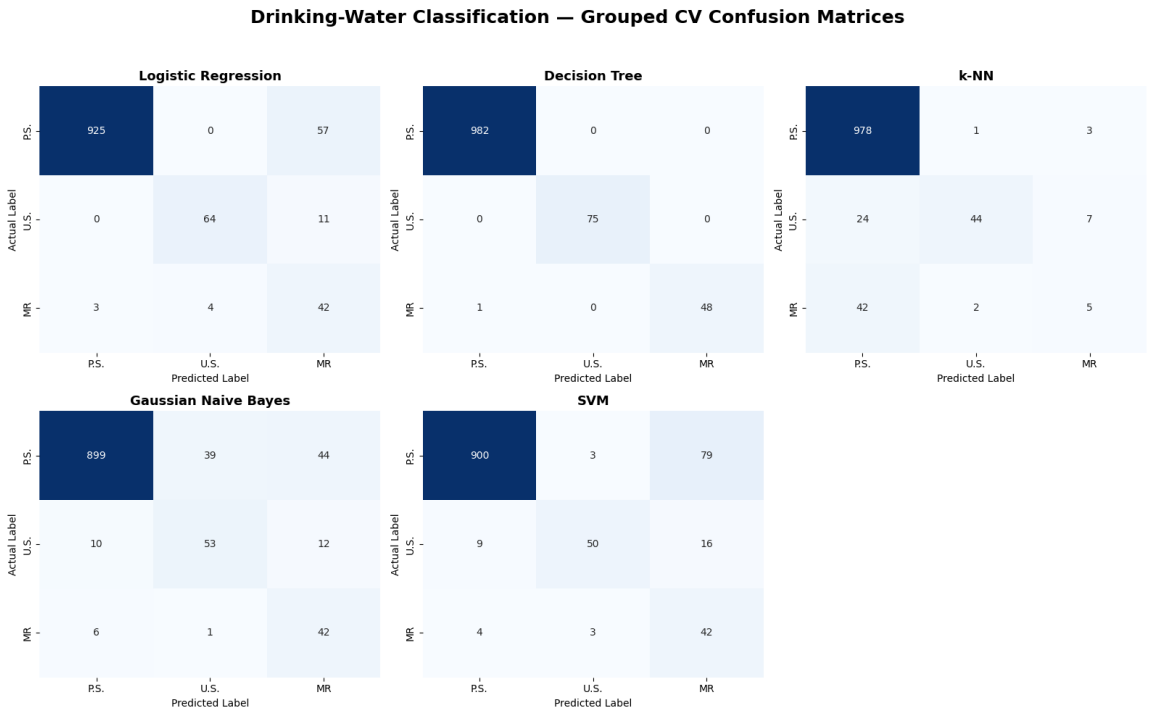



**Fig. 2.** [Insert figure: grouped-CV confusion matrices of the five classifiers (notebook cell 88).]

**Tuning and nested validation.** Table 5 gives the tuned settings and the nested, less optimistic estimates.

**Table 5.** RSC-based label: tuned grouped-CV scores and nested grouped-CV out-of-fold scores.

| Model | Selected parameters | Tuned macro-F1 (SD) | Tuned U.S. recall (SD) | Nested macro-F1 | Nested balanced acc. | Nested U.S. recall |
|---|---|---|---|---|---|---|
| Decision tree | no depth limit, min leaf 1, no class weight | 0.9963 (0.0074) | 1.0000 (0.0000) | 0.9964 | 0.9932 | 1.0000 |
| Logistic regression | C = 10, balanced | 0.8678 (0.0350) | 0.9277 (0.0441) | 0.8447 | 0.8607 | 0.9067 |
| SVM (RBF) | C = 10, γ = 0.01, balanced | 0.7798 (0.0162) | 0.8317 (0.0999) | 0.7314 | 0.7729 | 0.6533 |

Tuning improved logistic regression (macro-F1 from 0.799 to 0.845 nested) and the SVM (from 0.714 to 0.731 nested). Nested scores are lower than tuned scores for both, as expected.

**Why the tree is perfect.** The fitted tree splits on RSC at about 1.28 and 2.49 meq/L: RSC ≤ 1.28 gives P.S., values between 1.28 and 2.49 give MR, and values above 2.49 give U.S.; a further split on SO4 assigns P.S. on both sides and is redundant. Feature importance of the tree is 1.0 for RSC and 0 for all other features. Permutation importance on a held-out fold agrees (Table 6): RSC is the top feature for four of the five models, and SAR ranks first for k-NN.

**Table 6.** Permutation importance (drop in held-out macro-F1) on one grouped held-out fold, top two features per model.

| Model | First feature | Second feature |
|---|---|---|
| Decision tree | RSC (0.6256) | all others 0.0000 |
| Logistic regression | RSC (0.3603) | SAR (0.0904) |
| SVM (RBF) | RSC (0.1643) | SAR (0.1020) |
| Gaussian Naive Bayes | RSC (0.1410) | SAR (0.0744) |
| k-NN | SAR (0.1013) | RSC (0.0724) |

Because the class boundaries coincide with the published RSC thresholds, the near-perfect tree score reflects recovery of the labelling rule, not independent predictive skill from the water chemistry.

### 6.2 Irrigation (USSL) class

On the 13-label target, the SVM reached accuracy 0.9024 (998 of 1,106 records correct, 108 errors), macro-precision 0.4669, macro-recall (balanced accuracy) 0.4294, macro-F1 0.4422 and weighted F1 0.8928. The gap between accuracy and macro-F1 comes from the rare labels (Table 7): the three most frequent labels are predicted well, while five labels (C1S1, C2S2, C3S4, C4S3 and O.G) have recall 0. Macro-F1 on the five held-out folds ranged from 0.3837 to 0.6258 (mean 0.4982), showing that scores for rare classes depend strongly on which records fall in each fold.

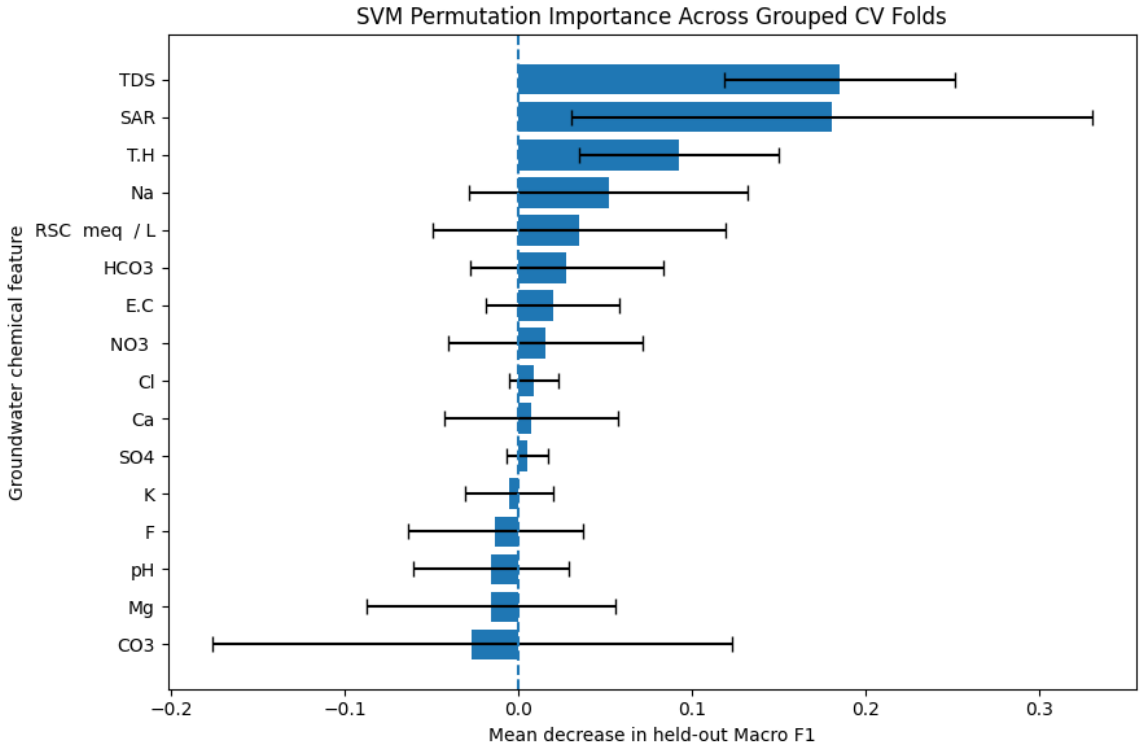
**Table 7.** Per-class results of the SVM on the 13-label irrigation class (pooled out-of-fold).

| Label | n | Precision | Recall | F1 |
|---|---|---|---|---|
| C3S1 | 696 | 0.9404 | 0.9756 | 0.9577 |
| C2S1 | 248 | 0.9388 | 0.9274 | 0.9331 |
| C4S1 | 87 | 0.7053 | 0.7701 | 0.7363 |
| C4S2 | 36 | 0.3182 | 0.1944 | 0.2414 |
| C3S2 | 12 | 0.6667 | 0.3333 | 0.4444 |
| C4S4 | 7 | 0.8333 | 0.7143 | 0.7692 |
| C3S3 | 6 | 0.6667 | 0.6667 | 0.6667 |
| C4S3 | 5 | 0.0000 | 0.0000 | 0.0000 |
| C1S1 | 3 | 0.0000 | 0.0000 | 0.0000 |
| O.G | 2 | 0.0000 | 0.0000 | 0.0000 |
| OG | 2 | 1.0000 | 1.0000 | 1.0000 |

| Label | n | Precision | Recall | F1 |
|---|---|---|---|---|
| C3S4 | 1 | 0.0000 | 0.0000 | 0.0000 |
| C2S2 | 1 | 0.0000 | 0.0000 | 0.0000 |

Fig. 3. SVM confusion matrix and class-wise recall for the 13-label irrigation (USSL) class.

Fig. 4. Permutation feature importance across five grouped folds for the irrigation-class SVM.




**Feature importance.** Across the five grouped folds, TDS (mean macro-F1 drop 0.1850) and SAR (0.1805) were the most important features, followed by T.H (0.0927), Na (0.0519) and RSC (0.0352). Several features had zero or negative importance (for example pH, Mg and CO3), meaning that shuffling them did not hurt performance. SAR and CO3 had large fold-to-fold standard deviations (about 0.15), so their rankings are unstable.

**Fig. 3.** [Insert figure: SVM confusion matrix and class-wise recall for the irrigation class (notebook cells 51/65).]

**Fig. 4.** [Insert figure: permutation importance across five folds (notebook cell 69).]

## 7 Discussion

**What worked.** For the common classes, simple models are sufficient: the SVM predicts C3S1 and C2S1 with F1 above 0.93, and logistic regression finds 85% of the unsuitable (U.S.) samples in the RSC-based task with precision 0.94. TDS and SAR matter most for the irrigation class, which fits the USSL definition,

in which the salinity class depends on conductivity and the sodium class on SAR [1]; we did not test how the labels were derived.

**Label derivation and leakage.** The RSC-based label is a function of RSC with thresholds at 1.25 and 2.5, and RSC itself is computed from four other features in the table (carbonate, bicarbonate, calcium and magnesium; Eq. 1). A model given RSC as an input is therefore solving a lookup problem, which explains why a depth-two tree is perfect while models that must also weigh 15 other features are not. This is a case of target leakage [7]. The tree's score should be reported as rule recovery. For a scientific claim about predicting suitability from water chemistry, RSC, SAR and the other derived indices should be removed from the inputs.

**Accuracy can mislead.** The majority-class share is 88.8% for the RSC-based label, so an SVM with 0.897 accuracy adds no accuracy over always predicting P.S., and k-NN's 0.929 accuracy hides a balanced accuracy of 0.563. For the irrigation class, accuracy of 0.902 contrasts with macro-F1 of 0.442.

**Limitations.** (i) Only three years of post-monsoon data are available. (ii) Grouping covers only identical chemical profiles, whereas most locations were sampled in more than one year, so the same site can appear in training and validation folds; spatial (district-wise) or temporal (year-wise) splits would be stricter. (iii) Eleven features are missing for 364 records, probably because of inconsistent column naming across the yearly files, and median imputation gives these records identical values. (iv) The 13-label target has very small classes and the O.G/OG spelling issue; scores for these labels are unreliable. (v) The irrigation task was evaluated with an SVM only, and tuning choices were partly made after inspecting earlier cross-validation results. (vi) No external test set was used.

**Ethics and SDG impact.** The data are aggregate environmental measurements with no personal information. The main risk is over-reliance: classifying unsuitable water as safe could harm crops, soils or people, so any screening tool should favour recall of the unsuitable class and be used only as a pre-screen for laboratory analysis. Reliable screening supports SDG 6 (target 6.3, water quality) and SDG 2 (sustainable agriculture).

## 8 Conclusion and Future Work

Using 1,106 groundwater records from Telangana and a grouped cross-validation protocol, we found that classical models reproduce common suitability classes well but struggle with rare ones. On the RSC-based label the decision tree's near-perfect scores (macro-F1 0.995) come from two thresholds on RSC, which is how the label is defined. On the 13-label irrigation class the SVM reached 0.902 accuracy but only 0.442 macro-F1. Future work will (i) repeat the experiments without RSC, SAR and other derived indices; (ii) merge the duplicate-named columns and the O.G/OG labels and group rare irrigation classes; (iii) use district-wise and year-wise splits; and (iv) compare tree ensembles and evaluate on new data.

## Declarations

**Data availability.** The three yearly CSV files were obtained from https://www.kaggle.com/datasets/sivapriyagarladinne/telangana-post-monsoon-ground-water-quality-data. Code and results: https://github.com/Taniya-roy20/Groundwater-quality-Telangana-/blob/main/Groundwater_Quality.ipynb

**Conflict of interest.** The author declares no conflict of interest.

**Funding.** None.

**Use of AI tools.**

Generative AI tools were extensively used throughout this project to support the research and development process. AI assistance was utilized for understanding the research problem, exploring the dataset, selecting and implementing suitable machine-learning models, developing and debugging code, interpreting model evaluation results, generating visualizations, and improving the structure and language of the research paper.

The AI-generated suggestions and outputs were reviewed and incorporated by the author as appropriate. The author acknowledges the use of AI tools and takes responsibility for reviewing the work, verifying the reported results, and ensuring the accuracy and integrity of the final submission.

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